# Yen-Nan (Aaron) Ho

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**AI Engineer**

Current Data Scientist bringing industrial signals, diagnostics, and LLM methods into verifiable product workflows. Lead product/domain architecture, model-assumption review, quality criteria, and acceptance, collaborating with engineers and AI agents on implementation, testing, integration, and delivery.

## Core Skills

- **Industrial AI/Research:** remaining useful life (RUL), phase-resolved partial discharge (PRPD)/vibration, FFT, anomaly detection, CNNs, ablations, confidence, and data-availability assessment.
- **AI Systems:** LangChain/LangGraph, report services, vLLM, OpenAI-compatible APIs, deterministic facts, fallbacks, and human validation.
- **Platform/Delivery:** Python, Django/Django Ninja, PostgreSQL, Redis, RabbitMQ, REST, async workflows, React/TypeScript, Docker, offline deployment, update/rollback, Git, and automated tests.

## Work Experience

### Data Scientist, Harbor Tech | Taipei | Nov 2025-Present

- **Product/domain architecture:** Led Harbor AI 2.0 product logic, translating partial discharge, RUL, trends, and data quality into equipment status and user flows. Traced a statistic to prevent sampling-point density being presented as discharge-event rate; corrected health calculations that included disconnected equipment. Engineering partners jointly integrated the Platform Core.
- **Model assumptions and validation:** Audited AI-generated RUL conclusions. Multi-parameter, multi-curve, and forward/reverse ablations exposed test constants misrepresented as observations, invalidating the conclusion. Connected research notebooks to production code to reduce drift. Separated fault type, severity, RUL, and low-confidence states without claiming unverified field accuracy.
- **Report-service productization:** Extracted reusable methods from field analysis, customer reports, and human validation. A July LangGraph POC separated deterministic facts from LLM narrative; August moved the report path into a platform-connected service; September integrated DOCX/frontend downloads into the mainline. Separately validated live vLLM capabilities. Field deployment, acceptance, and universal provider switching require further verification.
- **Platform and delivery:** Collaborated on Django async workflows, defining APIs, repository/artifact-by-reference contracts, task lifecycles, and validation boundaries. Integrated Docker/offline and cross-architecture installation, platform update/rollback, and auditable migration; led release integration and cut decisions.
- **Human-AI development environment:** Connected requirements, persistent context, tools, tests, records, and isolated branches. New requirements began with specifications and acceptance criteria; daily-test/monitoring anomalies became issues. AI agents supported research, implementation, and testing through both paths, submitting Bitbucket PRs for human review. I handled domain judgment and validation, sharing merge/release decisions with the reviewer.

- **End-of-line inspection:** Hosted customer workflow alignment and defined assets under test, measurement batches, decisions, reports, and acceptance mapping. Coordinated quality-testing handoff to the DQA team, test versions, and manuals. Later integration remains on a local branch outside the mainline; field calibration and customer acceptance remain open. Engineers own code implementation and fixes.
- **Reliability case:** In one Large-WAV release-test case, owned direction, assumption review, and validation while AI supported fixes. Peak memory fell from 35-42 GiB to 822 MiB; a 165-second failure became an approximately 48-second completion. Different inputs mean this is not an equal-workload benchmark or a system-wide performance claim.

### Data Science Intern, Institute of Sociology, Academia Sinica | Taipei | May 2019-Dec 2020

Used survey-area grouping, point-to-point distances, OR-Tools TSP, and Folium for field-survey routes, dispersion, and budget estimation.

## Selected Research and Projects

- **MCIENet, M.S. thesis (2024):** Independently led research decomposition, multi-scale CNNs, data pipelines, ablations, cross-dataset evaluation, and explainable AI (XAI) for DNA-sequence-based 3D chromatin interaction prediction. Adapted parts of ChINN preprocessing; retained cross-cell-line generalization limits.
- **Horus (2020):** Team-integrated multi-camera tracking, person ReID, and face recognition. Primarily owned ReID integration, similarity/dual-threshold decisions, and tracking-record workflows. First-place undergraduate capstone.
- **PropGMAN (2022):** A three-person team applied spatiotemporal modeling to regional land indicators from property transactions, retaining experiments and reproduction steps. Creative Award, AI for Asset Activation Innovation Contest.

## Research and Publications

- Dynamic traffic network representation model for improving the prediction performance of passenger flow. Knowledge-Based Systems, 2024.
- A Dynamic Embedding Method for Passenger Flow Estimation. IIAI AAI, 2021; Competitive Paper Award.
- Hierarchical Attention Network with Sentiment Lexicons for Stock Article Emotion Analysis. TCSE, 2020.

## Education

- National Chengchi University, M.S. in Computer Science | Sep 2021-Sep 2024.
- Soochow University, B.S. in Data Science | Sep 2017-Jul 2021.